

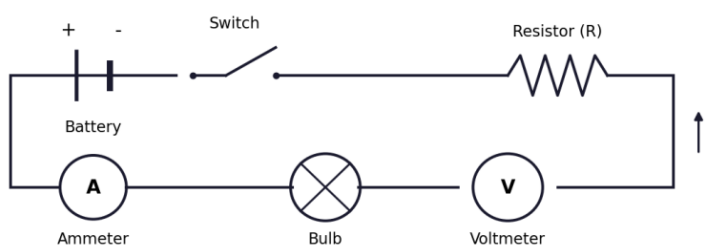
GradeSense - Answer Sheet

Name: Ananya Rao Roll No: 24B

Q1. Science (5 marks)

An electric circuit is a close path in which electric current can flow. The battery gives the potential difference which push the current around the circuit. Switch is used for open and close the circuit, if switch is close then current will flow and bulb will glow, if switch is open then no current flow through the circuit.

In the circuit the battery, switch, resistor, bulb and ammeter is connected in series because current pass through all of them one by one. Ammeter is connected in series because it measure the current flowing in circuit. Voltmeter is also connected in the circuit to measure the potential difference, as shown in diagram below -



When current pass through the circuit some of the current get used up by the bulb and resistor, that is why the bulb glow and become hot.

As per Ohms law, $V = IR$. So if we increase the resistance then the current flowing in the circuit will also increase, because more resistance need more current to push through it. If the resistance is less then the bulb will glow dim.